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Uplift deduced from remanent magnetization of a proterozoic basic dyke and the baked country rock in the Hoting area, Central Sweden: a palaeomagnetic and $^{40}Ar/^{39}Ar$ study

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SUMMARY

Remanent magnetization in dyke contact zones in the Hoting area of the Fennoscandian Shield in the central part of Sweden has been studied in order to establish the ambient temperature of the host rock and the depth of burial of the present erosion surface at the time of intrusion. A positive baked contact test for two Subjotnian dykes demonstrates the primary nature of the dyke magnetization. From the magnetic properties and the palaeomagnetic data, it can be concluded that the overprinting in the hybrid zone of one of the basic dykes is a partial thermoremanent magnetization. Reliable results were also obtained from a palaeointensity study of samples from the hybrid zone in the baked host rock. The study was performed in the laboratories at Luleå and at Scripps and a mean intensity of the Earth's magnetic field of 5.8 ± 1.9 μT was determined. The maximum temperature due to the dyke intrusion in the hybrid zone has been defined and from that an ambient temperature of 375 °C in the host rock at the time of intrusion has been calculated. This calculated temperature is not contradicted by the $^{40}Ar/^{39}Ar$ data. A palaeothermal gradient in the crust at *ca*. 1.6 Ga is calculated at *ca*. 34 °C km^{-1} and yields a depth of burial of the present erosion surface at *ca*. 10.4 km. This implies an uplift rate of 0.65 km (100 Ma)$^{-1}$. A slow cooling of the gabbroic host rock (*ca*. 3.5 °C Ma^{-1}) has been calculated from the difference in the U-Pb age of zircon (1.786 ± 0.010 Ga) and the $^{40}Ar/^{39}Ar$ biotite ages (1.648 ± 0.012 Ga; 1614 ± 0.024 Ga) of this study. This slow cooling resulted in a palaeomagnetic age of *ca*. 1.7 Ga for the gabbro, which is also the age of the determined palaeointensity. The calculated *ca*. 1.614 Ga palaeomagnetic pole from the basic dykes fulfils most of the criteria for a reliable pole and may be regarded as a new key-pole for Fennoscandia.

Key words: Palaeointensity; Palaeomagnetism applied to tectonics; Remagnetization.

INTRODUCTION

Analysis of the remanent magnetization in the host rock near an igneous intrusion may give us the temperature of the host rock at the time of intrusion (Schwarz 1977). The depth of burial of the present surface can be calculated from this temperature by dividing it by an estimated palaeothermal gradient. Such data, obtained for intrusions of different ages, may allow a quantitative evaluation of the history of vertical movement of the crust. Uplift studies based on remanent magnetization of diabase dyke contact zones have been carried out vastly in different parts of North America (Schwarz 1977; Buchan *et al.* 1980; Symons *et al.* 1980; Buchan & Schwarz 1981; Schwarz & Buchan 1982), but not from the Fennoscandian Shield; this paper is the first of its kind from Sweden.

THE PRINCIPLES OF UPLIFT STUDIES BASED ON ROCK MAGNETISM

When a dyke intrudes the country rock, the rock is heated within a certain distance from the contact, which is a function of the physical properties of the rocks, such as thermal diffusivity, specific heat, latent heat of crystallization and width of the dyke. Depending on the degree of heating and on the magnetic carriers of the host rock, the host rock will be remagnetized to various extents at an increasing distance from the dyke. If a remagnetization, or magnetic overprinting, occurs purely as thermoremanent magnetization (TRM) without accompanying chemical alteration, zones of complete, partial and no resetting are observed with increasing distance from the intrusion. Close to the contact, the original magnetization

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of the host rock will be completely destroyed and the reheated rock may acquire a magnetization with a direction parallel to the Earth's magnetic field at the time of intrusion. At a larger distance, the host rock will retain its original magnetization. Between these zones, there is a zone of partial resetting, or a hybrid zone, in which both magnetizations can be traced and the resultant remanence vector rotates from the direction of the dyke to the direction of the host rock. If the age difference between the host rock and the basic dyke or the apparent polar wander is large enough, the directions of the two magnetizations will be significantly different, and so it will be possible to identify them.

Since the younger component of magnetization in the hybrid zone is a partial TRM, the blocking temperature (T_b) spectra of the two components are essentially discrete and the component can be separated by stepwise heating. The maximum temperature (T_{max}) that the hybrid sample reached at the time of intrusion is the temperature at which the magnetization abruptly rotates from the younger dyke direction to the older host direction.

Buchan & Schwarz (1987) presented criteria to be fulfilled to ensure a thermoremanent origin of the hybrid magnetization: (1) there should be no systematic variation in the magnetic mineralogy at increasing distance from the dyke contact; (2) the dyke component in the hybrid zone should be isolated in blocking temperatures below that of the host component and (3) T_{max} should decrease systematically with distance from the dyke.

The increase in temperature due to the intrusion (ΔT_{max}) can be calculated theoretically on the basis of a 1-D heat-conducting model (Jaeger 1964), whereas the maximum temperature recorded in the hybrid zone (T_{max}) is obtained from vector analysis of the palaeomagnetic data. Subtracting ΔT_{max} from the maximum temperature (T_{max}) obtained in the vector analysis yields the ambient temperature (T_{amb}) of the host rock at the time of intrusion of the dyke. Thus,

$$T_{amb} = T_{max} - \Delta T_{max} \quad (1)$$

which is valid only in the hybrid zone where T_{max} falls within the total T_b spectrum.

To be able to apply this technique, the magnetization in the hybrid zone must be a pure TRM. Chemical changes may result in the acquisition of a chemical remanent magnetization (CRM), with a maximum unblocking temperature not related to T_{max}. The amount of time that the rock in the hybrid zone spent within 10 °C of T_{max} also plays an important role, since during this time some of the magnetic carriers with blocking temperatures higher than T_{max} may be unblocked and reset in the younger dyke direction. This viscous effect will cause a gradual change of the remanence direction from that of the dyke to that of the original host rock over a range of temperatures related to the T_b range of the rock, and so T_{max} is here defined from the vector plots as an average between the maximum temperature at which the overprinting remanence is defined and the minimum temperature at which the original host rock magnetization is preserved.

The estimated T_{max} must be corrected for the difference in cooling rates between the laboratory experiments and the original cooling of the dyke (Pullaiah *et al.* 1975); however, this correction is very small when T_{max} is close to the Curie temperature (Dodson & McClelland-Brown 1980). The ΔT_{max} increase at the distance x from the centre of a dyke of thickness $2d$ is calculated using the following expressions (Jaeger 1964), modified to allow $T_{amb} \neq 0$

$$\Delta T_{max} = [V + (L/c) - (T_{max} - \Delta T_{max})] \times 1/2 \left\{ \mathrm{erf}\left[(x+d)/2d\tau_{max}^{0.5}\right] - \mathrm{erf}\left[(x-d)/2d\tau_{max}^{0.5}\right] \right\} \quad (2)$$

where V is the magma temperature (taken at 1150 °C) at the time of intrusion ($t = 0$); L is the latent heat of crystallization (90 cal g^{-1} or 377 J g^{-1}); c is the specific heat of host and dyke, which in our case is approximately 0.28 cal g^{-1} deg. or 1.17 J g^{-1} deg.; τ_{max} is the maximum of the product of diffusivity and t, the elapsed time since the intrusion, divided by half the dyke width (d) squared. τ_{max} is also given by

$$\tau_{max} = (x/d)/\ln[(x+d)/(x-d)]. \quad (3)$$

The ambient temperature is then calculated from the expression (1), where T_{max} has been corrected for the viscosity effect using the blocking temperature curves for magnetite of Pullaiah *et al.* (1975).

A necessary condition for eq. (2) is that the distance of a hybrid sample from the contact of the dyke is larger than $1/4d$ (Jaeger 1964). These relations can only be used when the heat transfer is through conduction and the effect of water circulation and volatiles escaping from the dyke can be neglected. In permeable rocks, the latter factors may significantly decrease the ΔT_{max} (Buchan *et al.* 1980), but in our case of crystalline rocks, these effects are not considered.

Numerical solutions on the problem of dykes cooling by conduction have been presented by Delaney (1987, 1988). These solutions are particularly useful for calculations close to the contact of the dyke.

The depth of burial is then determined by dividing the ambient temperature by the palaeo-thermal gradient in the crust. The gradient is estimated from the present thermal gradient and the change in heat production of the rock due to the decay of radioactive minerals. This can only be a rough estimation and so it represents the main source of error when determining the depth of burial.

GEOLOGY

The Fennoscandian Shield can be divided into tectonomagmatic blocks representing a geochronological zonation and accretionary growth during successive orogenic events, i.e. the Loopian (2.9–2.6 Ga), the Svecofennian (1.9–1.85 Ga), the Gothian and the Transscandinavian Igneous Belt (TIB; 1.75–1.5 Ga), the Sveconorwegian (1.25–0.85 Ga) and the Caledonian (0.6–0.4 Ga), the oldest parts being in the northeast and the youngest in the southwest. The Svecofennian (2.0–1.75 Ga) province is divided into northern, central and southern subprovinces (Gaál & Gorbatschev 1987). Dykes of various generations intrude the Svecofennian province and in this study we sampled basic dykes and a gabbro formation in the Central Svecofennian Subprovince (Fig. 1).

The Hoting gabbro (64.15°N, 16.28°E) is located in the western part of the Central Svecofennian Subprovince and it is bound by the Revsund granodiorite to granite and by metasediments (Fig. 1). The granite has a U-Pb age from zircon of 1.782 ± 0.012 Ga (Hellström & Hegardt 2003) and a U-Pb zircon dating of the gabbro reveals an age of 1.786 ± 0.010 Ga (Hellström & Larson 2003), which means the emplacement of the gabbro was contemporaneous with that of the Revsund granite. This is further supported by mingling structures between the granite and the gabbro. There are a number of NE–ENE trending basic dykes in the region and some of these cut the gabbro. The width of the dykes is 1.0–1.5 m and they are generally vertical (Moakhar & Elming 1998). In the south and southeast of the gabbro, small rapakivi complexes have intruded into the Revsund granite. These intrusions form part of a discontinous belt of anorogenic plutons running from western Russia over southern Finland, Estonia and Latvia to central Sweden. The rapakivis in central northern Sweden are *ca*. 1.58–1.51 Ga old (Andersson 1997).

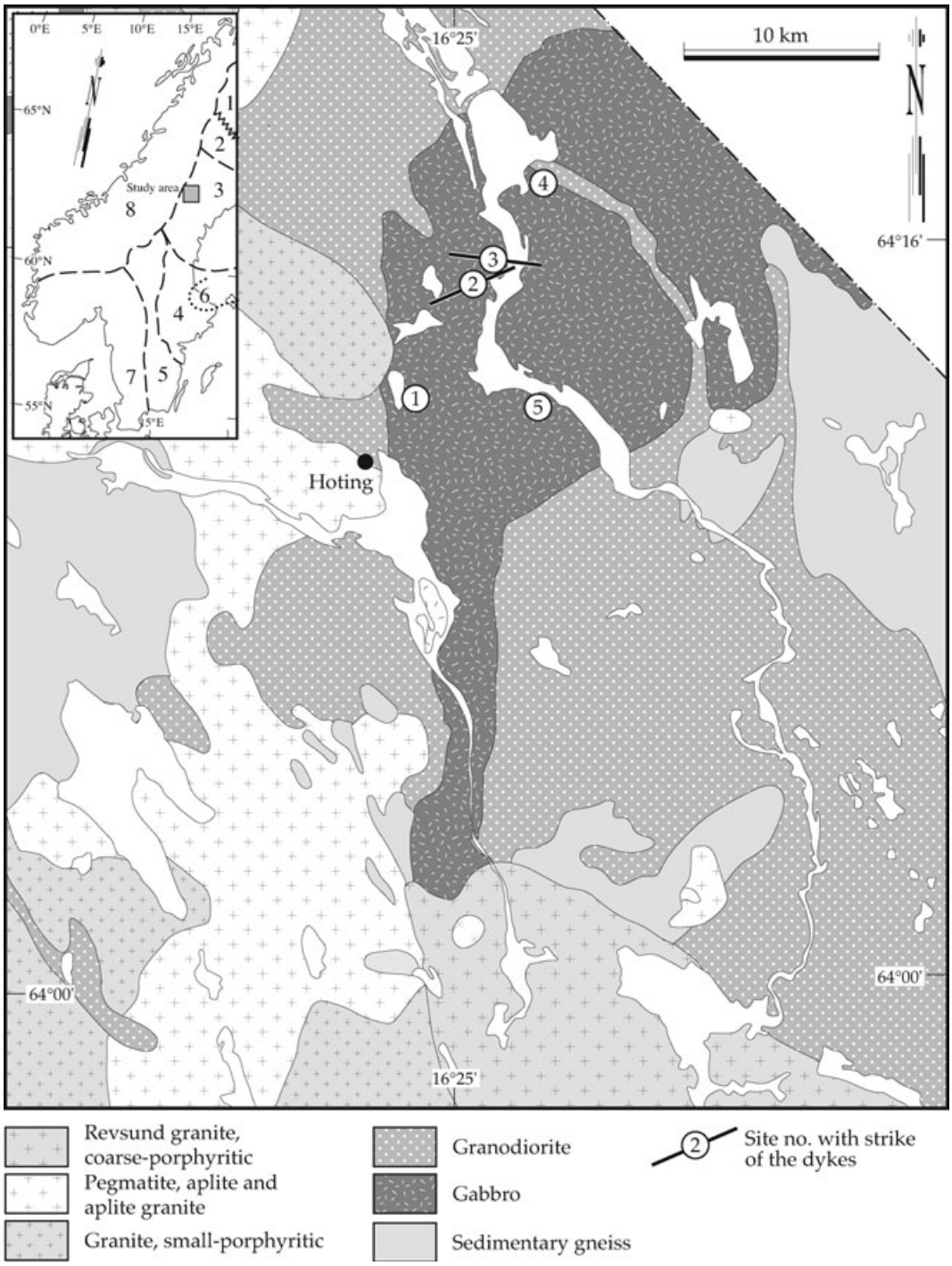


Figure 1. Geological map and location of the sampling sites in the Hoting area, central Sweden. The inset figure shows the location of the study area and a tectonomagmatic block division of the Fennoscandian Shield (modified after Elming *et al.* 1993). 1, Northern Svecofennian subprovince; 2, Skellefte belt; 3, Central Svecofennian subprovince; 4, Southern Svecofennian subprovince; 5, Transscandinavian igneous belt; 6, Rapakivi granites; 7, Southwest Scandinavian province and 8, Caledonides.

A palaeomagnetic and geochemical study has been performed on the basic dyke swarm (Elming *et al.* 2004). Different palaeomagnetic signatures and chemical compositions indicate that this swarm is composed of two generations of dykes, one of which is probably related to rapakivi magmatism.

SAMPLING AND LABORATORY TECHNIQUE

Some 100 samples have been collected from 2 dykes and the gabbro (Fig. 1). The samples were taken with a portable drill and oriented

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with the sun and magnetic compasses. The samples were collected from various parts of the dykes and from the adjacent country rock along profiles running roughly perpendicular to the contact of the dykes. The profiles were extended far away from the contact to define the magnetization of the gabbro, not influenced by the heat from the dyke intrusion. The gabbro was also sampled at sites located far away from the dykes. The core samples were cut into at least two specimens and the specimens were demagnetized conventionally by thermal (Schonstedt, TSD1) and alternating field (AF) techniques. The temperature during the thermal demagnetization has been carefully calibrated by measuring the temperature inside the specimens at various positions in the oven during thermal demagnetization. The temperature of the specimens has in this way been determined with an accuracy of better than ±5 °C. The specimens were demagnetized in steps of 5 and 10 mT up to 100 mT AF in an attempt to delineate zones of contact, hybrid and host magnetization. When successful, specimens from each zone were thermally demagnetized to a maximum of 600 °C with intervals of 50 °C up to 500 and 10 °C between 500 and 600 °C. Palaeointensities from seven specimens were measured at Luleå with the Coe method (Coe 1967). At Scripps, the new IZZI method (Tauxe & Staudigel 2004) was used on specimens from two successful samples measured at Luleå University. The specimens were sliced in order to obtain cylinders of about 2 mm height and, in total, nine specimens were measured.

The measurements of the remanent magnetization were performed with a spinner (AGICO, JR5) and a SQUID (2G-DC) magnetometer. The remanence vectors were defined by least square fit of vectors in 3-D space (Torsvik *et al.* 1996) and by principle component analyses (Kirschvink 1980). To check for magnetochemical changes in the host rock, bulk susceptibility and susceptibility-temperature variations of samples from the gabbro were measured with AGICO, KLY-3 and CS-3 equipment.

^{40}Ar/^{39}Ar AGE DETERMINATIONS

As with magnetism, the argon isotopic system in country rock minerals can be reset or partially reset due to the intrusion of a dyke (e.g. Hyodo *et al.* 1993; McDougall & Harrison 1999). The amount of resetting (open system behaviour) is a function of the temperature and size of the intruding body, the ambient temperature of the country rock and the mineral.

^{40}Ar/^{39}Ar age determinations are here presented for the gabbro at sites 1 and 3 and for the cross-cutting basic dyke at site 3 (Fig. 1). Palaeomagnetic core samples were submitted for ^{40}Ar/^{39}Ar analysis at the Geochronology Laboratory at the University of Alaska Fairbanks. Biotite and plagioclase grains were extracted for mineral dating and, for the whole-rock dating, the cores were crushed, washed and sieved to obtain small (∼0.5 mm) chips for dating. The monitor mineral MMhb-1 (Samson & Alexander 1987) with an age of 513.9 Ma (Lanphere & Dalrymple 2000) was used to monitor neutron flux (and calculate the irradiation parameter, J). The samples and standards were wrapped in aluminum foil and loaded into aluminum cans of 2.5 cm diameter and 6 cm height. The samples were irradiated in position 5c of the uranium-enriched research reactor of McMaster University in Hamilton, Ontario, Canada for 20 megawatt hours.

Upon their return from the reactor, the sample and monitors were loaded into 2 mm diameter holes in a copper tray that was then loaded into an ultra-high vacuum extraction line. The monitors were fused, and samples heated in a stepwise manner, using a 6-watt argon–ion laser following the technique described in York *et al.* (1981), Layer *et al.* (1987) and Layer (2000). Argon purification was achieved using a liquid nitrogen cold trap and a SAES Zr-Al getter at 400 °C. The samples were analysed in a VG-3600 mass spectrometer at the Geophysical Institute, University of Alaska Fairbanks. The argon isotopes measured were corrected for system blank and mass discrimination, as well as calcium, potassium and chlorine interference reactions following procedures outlined in McDougall & Harrison (1999). Plateau ages were based on the criteria of three or more contiguous fractions (with a roughly constant Ca/K ratio) accounting for more than 50 per cent gas release, having a Mean Square Weighted Deviation (MSDV) less than 2.5. The analytical data is given in the Appendix and a summary of all the ^{40}Ar/^{39}Ar results is given in Table 1, with all ages quoted to the +/– 1σ level and calculated using the constants of Steiger & Jaeger (1977). Integrated and plateau ages include error in the irradiation parameter (0.5 per cent).

Biotite from the gabbro at a site far away from the dykes was dated with a good plateau age at 1.648 ± 0.012 Ga (Fig. 2 and Table 1). A whole-rock sample was dated from the basic dyke revealing a plateau age of 1.366 ± 0.035 Ga. The baked gabbro in contact with the dyke was also dated and biotite, plagioclase and whole-rock ages were obtained at 1.614 ± 0.024, 0.892 ± 0.302 and 0.840 ± 0.150 Ga, respectively (Fig. 2).

Table 1. ^{40}Ar/^{39}Ar mineral and whole-rock ages of the gabbro in the Hoting area, the baked gabbro and the basic dyke at site 3 (for analytical data see the Appendix). Note that the whole-rock age for the basic dyke is probably a reset age.

Site	Rock	Min	Integrated age (Ma)	Plateau age (Ma)	Plateau information	Other comments
Site 1	Gabbro	Biotite	1601 ± 12	1648 ± 12	5 fractions 84 per cent 39Ar release MSWD = 1.8	Good plateau
Site 3	Baked gabbro	Biotite	1564 ± 34	1614 ± 24	5 fractions 89 per cent 39Ar release MSWD = 0.4	Good plateau
		Plagioclase	646 ± 233	892 ± 302	5 fractions 52 per cent 39Ar release MSWD = 0.6	Low K content Poor precision
		Whole rock	646 ± 141	840 ± 150	8 fractions 77 per cent 39Ar release	Low K content Reset age?
	Basic dyke	Whole rock	1293 ± 42	1366 ± 35	5 fractions 81 per cent 39Ar release MSWD = 1.6	Good plateau ??

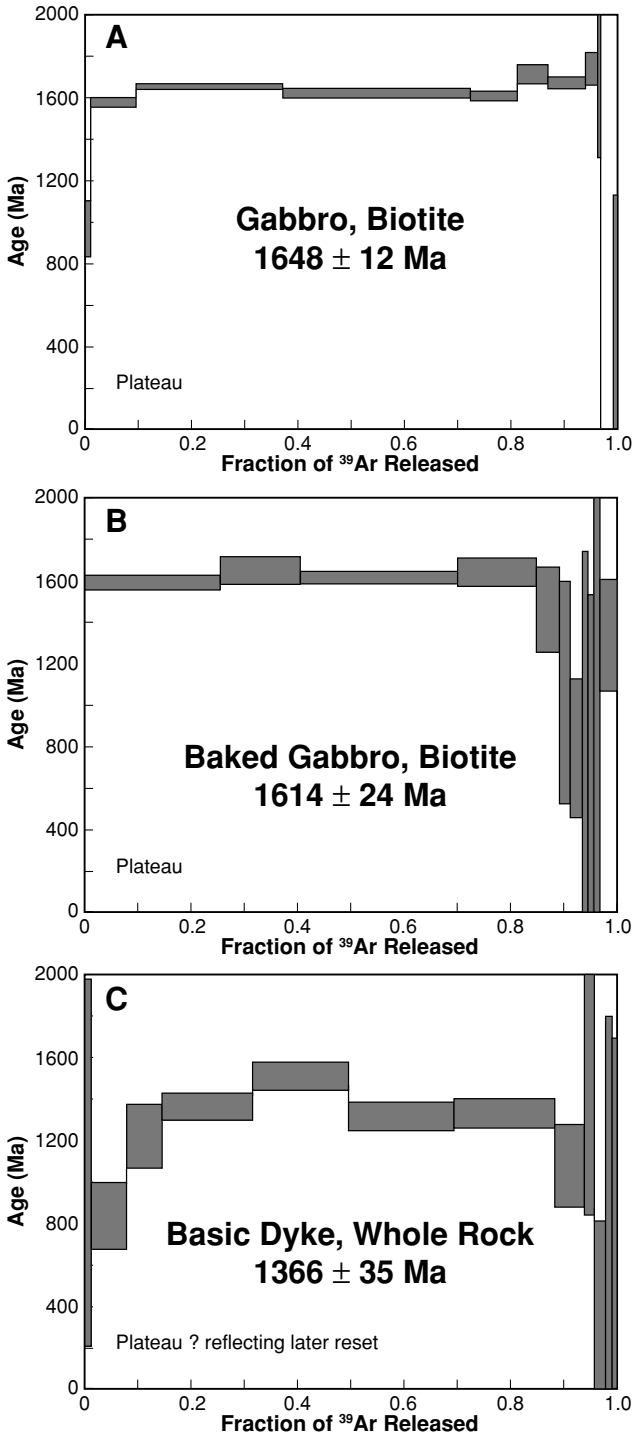


Figure 2. $^{40}Ar/^{39}Ar$ age spectra from mineral (biotite) and whole-rock samples. The whole rock plateau age 1.366 ± 0.035 Ga of the basic dyke is probably a reset age, while diffusion modelling gives an age of *ca*. 1.6 Ga. The samples are collected from the palaeomagnetic drilling cores. See text for analytical procedures.

The whole-rock sample from the basic dyke has a complex Ca/K spectrum, reflecting the fact that different phases have been dated. The first *ca*. 50 per cent has Ca/K values <3 and increases stepwise up to *ca*. 1.5 Ga. Diffusion modelling of this part of the spectrum gives an original age of *ca*. 1.6 Ga, with a significant Ar loss (>50 per cent) at about 0.6–0.8 Ga. The second part of the release has higher Ca/K values (>6) and probably reflects dating of the fine-grained plagioclase in this sample. These ages go down to the plagioclase age of the baked host rock. Thus, it is possible that the basic dyke and the baked gabbro were affected by a low temperature event in the late Proterozoic/early Palaeozoic. The biotite in the host rock escaped this event.

PALAEOMAGNETIC RESULTS

The gabbro carries an NRM of fairly high coercivity with a median destructive field (MDF; Dunlop & Özdemir 1997) of *ca*. 15 mT. The characteristic remanent magnetization (ChRM) is isolated in high coercivity (Hc) and unblocking temperature (T_{ub}) ranges (Figs 3a and b). The directions are well defined and a mean direction of five site means is calculated at Dec. = 331.9° and Inc. = 37.5° (Table 2). Thermal demagnetizations (Fig. 3) indicate that magnetite is the principle magnetic carrier.

The T_{ub} spectrum of the NRM in the dyke at site 2 is rather wide (Fig. 3c) and the coercivity is low with MDF of *ca*. 7 mT (Fig. 3d); however, a ChRM is isolated in T_{ub} > 530 °C with a site mean of Dec. = 53.8°, Inc. = −9.8° (Fig. 4 and Table 2).

The dyke width is 0.7–1 m and its exposed length is *ca*. 2 m. Samples were collected along a profile perpendicular to the contact and also at different positions along the dyke near to the contacts (Fig. 4).

In the actual contact to the dyke, the baked gabbro carries a direction similar to that of the dyke. At increasing distance, the high unblocking temperature ($T_{ub} \geq 500$ °C) directions change towards a direction of opposite polarity to that of the gabbro and finally to a direction of normal polarity (Fig. 4). This suggests that the gabbro here carries a magnetization of two polarities and that the dyke intruded in a part of the gabbro with a remanence direction opposite to the predominant direction. For many of the samples collected close to the dyke, the high T_{ub} remanence direction falls in a great circle between the remanence direction of the dyke and the gabbro (Fig. 4), indicating overlapping T_{ub} spectra of the two magnetizations. From the unbaked gabbro to the contact of the dyke, there is a strong increase in magnetic susceptibility (Fig. 5), suggesting that there has been a chemical alteration of the gabbro. Since the area of reversed gabbro directions coincide with the increase in magnetic susceptibility it may also be possible that the reversed directions are result of thermochemical alteration caused by the dyke intrusion. A chemical alteration would explain the overlapping T_{ub} spectra and the problem of separating the dyke from the gabbro component of magnetization in the area ≤ *ca*. 1.5 m from the contact of the dyke.

The dyke at site 3 is 1.35 m wide, vertical and trends 95 °E. Stable remanent magnetizations are defined in $T_{ub} \geq 530$ °C and in AF ≥ 15 mT (Figs 3e and f). The direction of remanence varies between different parts of the dyke. The mean direction of the samples from the margin of the dyke or near to it (Dec. = 67.2°, Inc. = 11.0°; Table 2) is different from the direction in the samples, which were collected from the central part of the dyke (Dec. = 40.0°, Inc. = 26.9°; Table 2 and Fig. 6).

Initial AF demagnetizations of specimens from the baked gabbro show four different zones of magnetization along a profile from the contact to the unbaked gabbro. Closest to the contact of the dyke, the gabbro carries low coercivity magnetizations with directions similar to that of the dyke margin and, at higher coercivities, directions that are mixed (Fig. 7). With increasing distance from the dyke, the directions change gradually from that of the dyke margin to that

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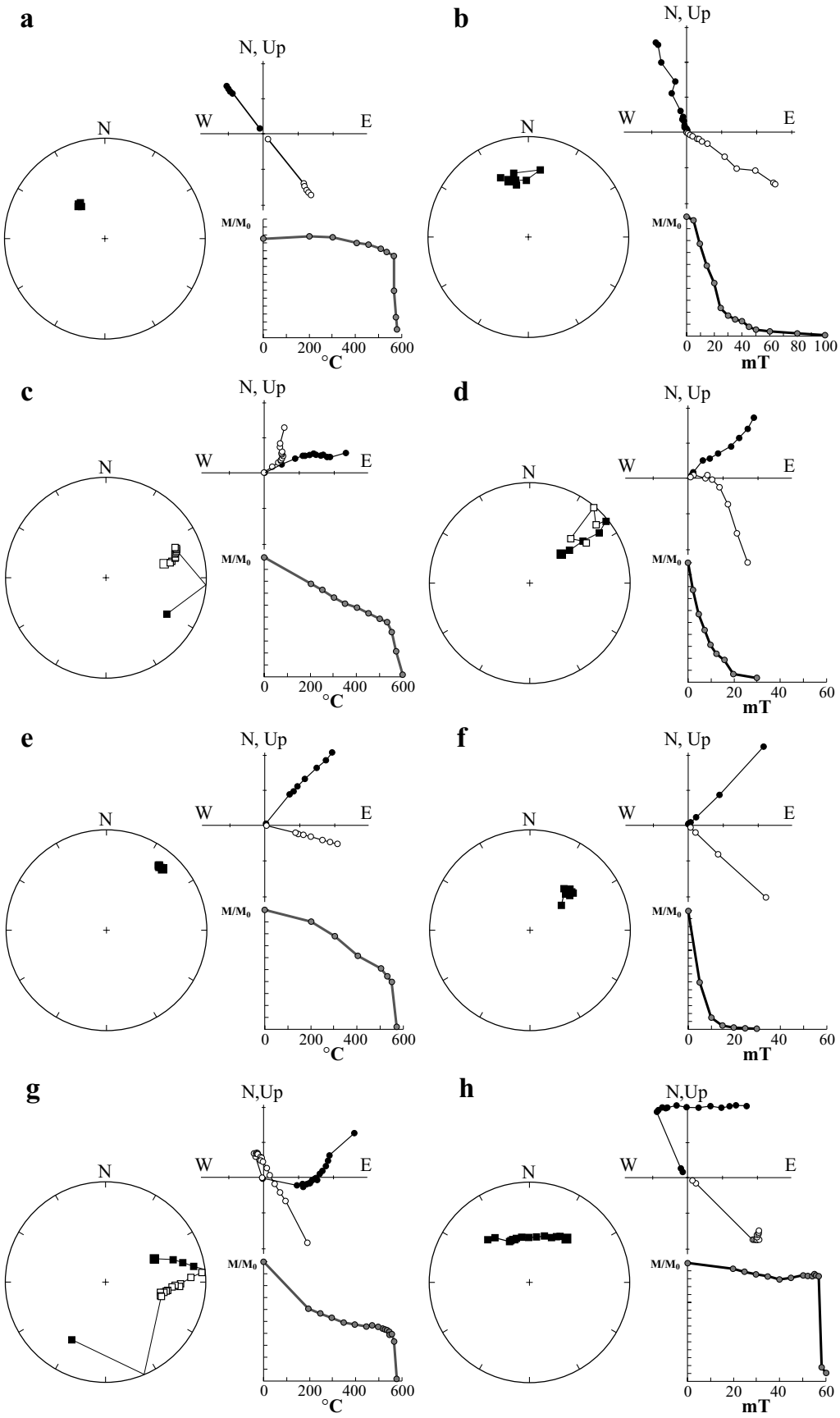


Figure 3. Examples of remanence behaviour during thermal and AF demagnetizations of samples from the gabbro (a and b), basic dyke at site 2 (c and d), basic dyke at site 3 (e and f), baked gabbro at site 2 (g) and at site 3 (h).

Table 2. Palaeomagnetic data from the gabbro, the basic dykes and the baked gabbro at site 2 and site 3.

Site	Glat (N°)/Glon (E°)	Rock	*N*/*n*	Dec. (°)	Inc. (°)	α_{95} (°)	*k*	Plat. (N°)	Plon. (E°)	A_{95} (°)
1	64°07.71′/16° 13.64′	Gabbro	1/7*	315.5	39.4	7.5	65	39.0	252.8	7.0
2	64°09.30′/16°15.86′	Gabbro, N**	1/13*	341.5	28.8	6.9	37	39.5	219.7	5.6
		Gabbro, R**	1/10*	139.9	−53.6	7.4	44	51.3	254.9	8.6
		Baked gabbro	1/7*	89.4	−6.7	25.3	7	−2.8	108.3	18.0
		Basic dyke*	1/8*	53.8	−9.8	12.1	22	10.3	141.5	8.7
3	64°09.89′/16°16.36′	Gabbro, H**	1/10*	339.3	44.9	3.6	178	50.0	225.8	3.6
		Baked Gabbro H*	1/21*	85.4	3.0	4.7	46	3.4	109.8	3.6
		Basic dyke Central part	1/12*	40.0	26.9	7.1	39	33.0	148.3	5.7
		Basic dyke margin	1/12*	67.2	11.0	10.3	19	14.8	124.7	7.4
		Mean Basic dyke*	1/24*	54.1	19.6	8.3	14	24.1	135.4	6.3
4	64°10.81′/16°18.15′	Gabbro**	1/7*	319.3	42.3	10.9	32	42.4	249.7	10.5
5	64°07.73′/16°18.27′	Gabbro**	1/10*	341.3	29.1	7.1	62	39.8	220.0	5.8
		Mean Gabbro**	5*/47	331.9	37.5	12.1	41	43.0	233.3	10.9
		Mean Basic dyke*	4*/41	45.1	8.5	19.1	24	21.9	146.7	13.7

Notes: Glat., Glon. is the geographical latitude and longitude of the sampling site. *N* is the number of sites and *n* refers to the number of samples. *denotes the level of the mean calculation. Dec., Inc., is the mean declination and inclination, respectively. α_{95} is the radius of the cone of confidence at a 95 per cent probability level. Plat., Plon., denotes the latitude and longitude of the palaeomagnetic pole and A_{95} is the radius of the 95 per cent confidence circle of the pole (Irving 1964). The host rock gabbro directions in site 3 forming the mean (*H*) are isolated in the hybrid zone at temperatures ≥560 °C. The mean basic dyke direction is calculated from the mean of sites 2 and 3 combined with the mean of sites 20a and 21 in Elming *et al.* (2004; their table 1).

of the host rock. Within a distance of *ca*. 1.15 m from the dyke contact, the two components can not be separated, the resulting magnetization being a mixture of the two, while at distances from 1.3 to 2.3 m, in the hybrid zone, the magnetization vectors of the dyke and the unbaked gabbro can be isolated (Fig. 7). The dyke directions are isolated in low Hc ranges, while the host rock ones are identified in high Hc ranges.

Thermal demagnetizations were then applied to reconstruct the effect of thermal remagnetization. The direction of remanence from samples in the chilled margin of the dyke is also identified in the gabbro close to the contact and in the hybrid zone (Fig. 6).

ORIGIN OF MAGNETIZATION AT SITE 3

For the technique to work, there must be no chemical alteration of the rock either in nature or during the laboratory experiments, as we can see in the samples from site 2.

The thermal demagnetizations of samples from the different zones at site 3 reflect differences in T_{ub} spectra. Samples from the hybrid zone retain their magnetization to very high temperatures and lose most of it in a narrow temperature range close to the Curie temperature of magnetite (Fig. 8a), while samples close to the contact gradually lose a large part of the magnetization with increasing temperatures. The unbaked gabbro also retains most of its magnetization up to 520 °C (=90 per cent) and the intensity decreases within a temperature range similar to that of the hybrid samples.

From AF demagnetizations, differences in coercivity between gabbro samples from the contact zone and the hybrid zone can be observed (Fig. 8b). In the contact zone, a large portion of the remanence is soft, while in the hybrid zone, the coercivity is higher and not different from that of the host rock.

Susceptibility-temperature measurements of the unbaked gabbro in site 3 do not show any signs of magnetochemical changes during heating up to 565 °C, which is the maximum temperature (T_{max}) at which the gabbro in the hybrid zone is reheated. At temperatures higher than 600 °C, irreversible changes of susceptibility are observed (Fig. 8c). This means that the magnetization in the hybrid zone is most probably of TRM origin, while there has probably been a chemical alteration in the contact zone.

PALAEOINTENSITY

With reference to the good arguments for a TRM origin of the remanence, a preliminary palaeointensity study was performed both at the Luleå University of Technology and at the Scripps Institution of Oceanography.

At Luleå, reliable results were obtained from two of the seven specimens collected in the hybrid zone at site 3. This is expressed by the high quality factors q (Coe *et al.* 1978) of 35.9 and 6.6, from specimens collected at 1.47 m (HT06–05) and 1.65 m (HT07–05) from the contact with the dyke, respectively. Plotting the pNRM lost versus pTRM gained in an Arai plot (Nagata *et al.* 1963) for the specimen collected at 1.47 m, the graph is perfectly linear in the temperature interval 525–580 °C (Fig. 9). Also, from the other sample in the hybrid zone, the results demonstrate an ideal behaviour and the graph is very close to linear. The linear behaviour is shown in the temperature range of *ca*. 525–580 °C, that is, in the range where we can see the shift from the young dyke direction and the original gabbro direction. This interval also corresponds to the main unblocking interval of magnetization, and so we observe high f factors (Coe 1978; 0.49 and 0.60, respectively) for each palaeointensity determination. The palaeointensity determined from the two samples in Luleå is 10.28 and 7.08 μT, respectively, using a laboratory field of 50 μT (Table 3).

Based on the low palaeointensities obtained from the Luleå experiments, at Scripps it was decided to apply a 15 μT field, following the recommendations after Selkin *et al.* (2007).

The main problem with the experiments performed at Scripps resides in the fact that there is a tendency for the specimen to alter, as can be observed from the relatively high—between 12 and 32 per cent—DRATS parameters (Tauxe & Staudigel 2004; Table 3). In their approach, Tauxe & Staudigel (2004) studied the cumulative distribution functions of each parameter to decide the threshold value for each selection criteria. Doing this on a set of nine specimens is not straightforward, so we take a maximum DRATS

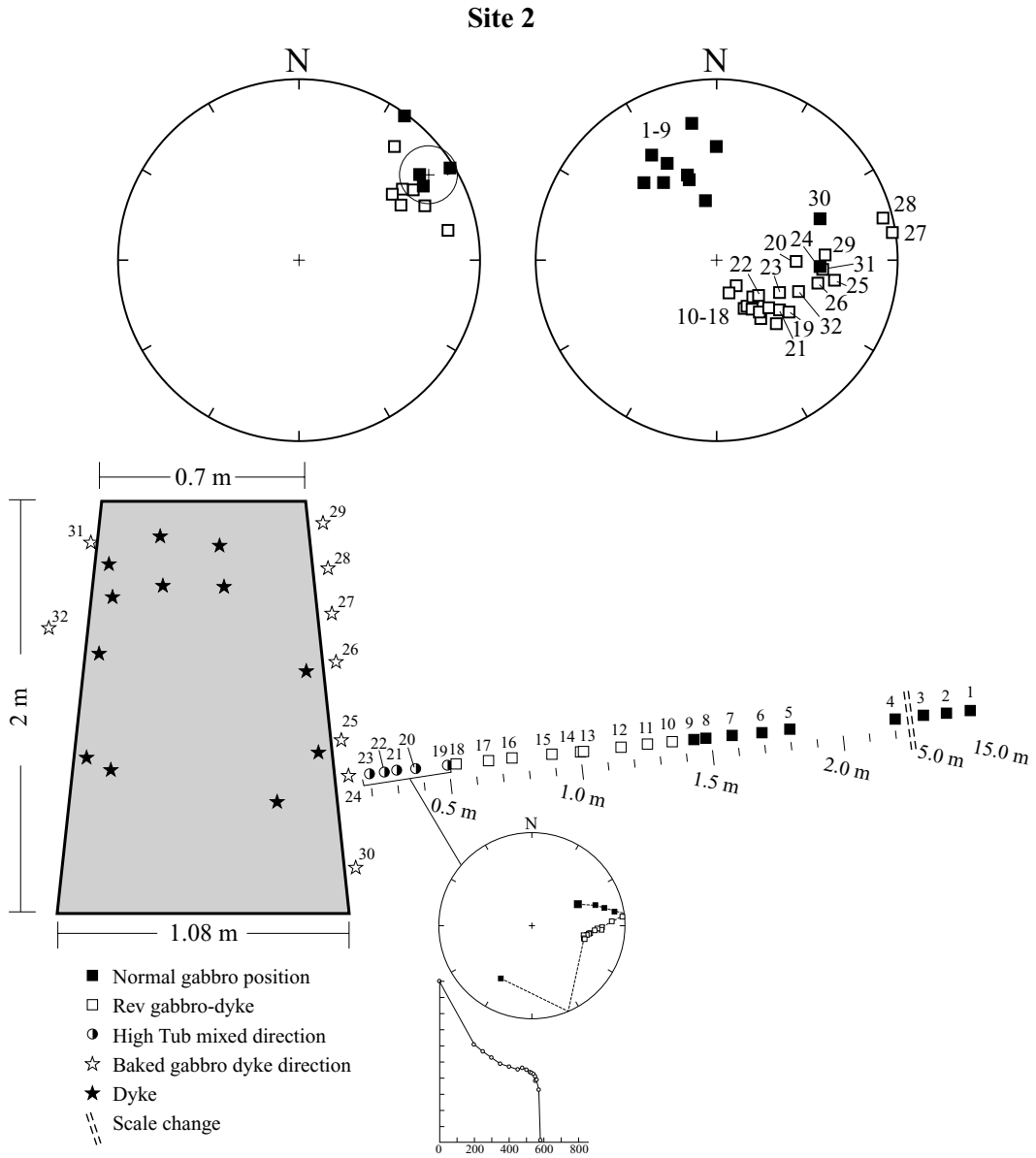


Figure 4. The positions of the samples collected from the basic dyke and the host rock at site 2. On top of the basic dyke are the directions of the ChRM plotted in a stereographic projection and the corresponding directions defined in the host rock are plotted on top of the profile of the host rock sampling. Open symbols denote negative inclinations and the ellipse shows the 95 per cent cone of confidence of the mean direction. Note the gradual change in ChRM directions of the host rock with increasing distance from the dyke from that of the basic dyke over a reversed direction of the gabbro to a normal direction of the gabbro.

value of 20 per cent. The selected dataset consists of five measurements performed at Scripps, and yields a mean intensity of 4.9 ± 1.3 μT (Table 3).

The two measurements performed at Luleå show higher palaeointensity values compared with the ones measured at Scripps using a lower laboratory field. However, in both experiments, the calculated palaeointensities are low and in terms of amplitude not very different. Taking a mean of all measurements, both from Luleå and Scripps, a mean palaeointensity is obtained at 5.8 ± 1.9 μT.

From the calculated mean intensities of all measurements, the mean of the measurements at Scripps and the mean of those with DRATS <20 per cent, the Virtual Dipole Moments (VDM) are determined at 1.0 ± 0.32 × 10^{22}, 0.88 ± 0.23 × 10^{22} and 0.90 ± 0.19 10 × 10^{22} Am^2, respectively (Table 3). The VDMs are thus not significantly different and here we use the mean of all measurements, determined at two different laboratories, to represent the VDM at the time of magnetization of the gabbro.

ESTIMATION OF THE MAXIMUM TEMPERATURE (T_{max}) IN THE HYBRID ZONE OF SITE 3

Since the rock in the hybrid zone at site 3 does not show signs of chemical alteration and the remanence is indicated to be a TRM, while the host rock in site 2 seems chemically altered, estimation of maximum temperature from the remanent magnetization is thus restricted to site 3.

The direction of magnetization isolated in the baked contact and hybrid samples is the same as that in the marginal parts of the dyke

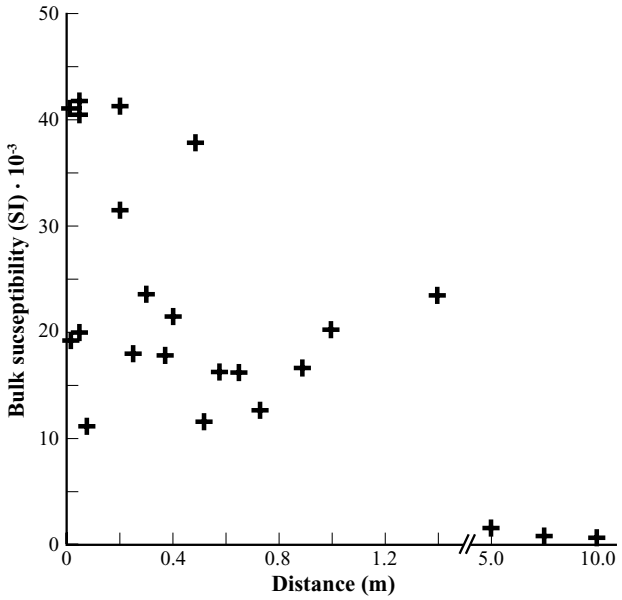


Figure 5. Magnetic susceptibility of samples collected at various distances from the contact to the basic dyke at site 2. Note the increase and big variation in susceptibility when approaching the dyke.

(Figs 6 and 10; Table 2). The baked zone extends to a distance of *ca*. 1.15 m from the contact. This is followed by a hybrid zone between 1.30 and 2.3 m, where NRM directions gradually swing from that of the dyke to that of the unbaked host rock. The pure host rock direction is present at a distance of 20 m, although it is also observed as an end point on stepwise AF and thermal demagnetizations of most hybrid samples. Through the hybrid zone (Fig. 10), the NRM directions move progressively away from the dyke direction towards a north-northwesterly direction, which represents the host magnetization. During thermal demagnetization, the directions of hybrid samples swing smoothly along great circles towards the host direction. The mean host rock remanence direction in the hybrid zone at site 3 (Dec. = 339.3°, Inc. = 44.9°, $\alpha_{95} = 3.6°$; Table 2) is similar to the site mean direction defined at other sites of the gabbro (Table 2).

If the T_{ub} spectrum of the gabbro host rock direction and that of the baked gabbro do not significantly overlap, as should occur in the case for purely thermal overprinting without viscous effects, then two straight-line segments are produced in the vector plot. The intersection of these lines defines T_{max}, the maximum T_{ub} of the magnetization that overprinted the original magnetization of the host rock. Viscous effects may cause some overlap, resulting in a more gradual transition from one straight line to another (Fig. 10). Chemical overprinting is unlikely to yield a smooth directional change in a continuous T_{ub} spectrum. In our case, there is a more gradual change in directions indicating viscous effects. T_{max} is therefore determined by averaging the two temperatures that define the beginning and the end of the viscous change in direction, that is, the maximum temperature at which the dyke direction of magnetization is identified and the minimum temperature at which the original gabbro direction is identified (Table 4). To correct T_{max} for viscous effects due to unblocking of those grains which were at or near the maximum temperature (T_{max}) for an interval of time, the standard curves of Pullaiah *et al.* (1975) have been used. The time (t) that the samples spent at maximum temperature in the laboratory is

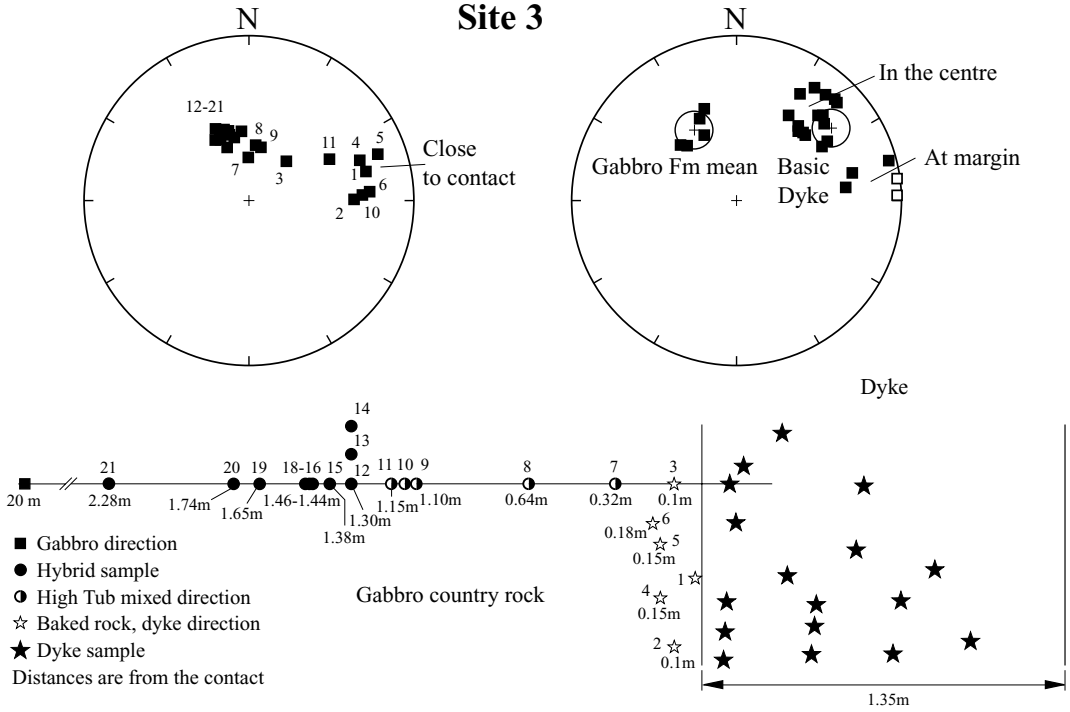


Figure 6. The position of samples collected from the basic dyke and the host rock at site 3. The distance from the dyke is given in metres. Host rock samples, in which pure dyke directions, mixed directions and two separable directions (hybrid zone) are identified, are marked with different symbols. The gradual change in ChRM directions with increasing distance from the dyke contact is shown in the stereographic plot on top of the host rock sampling profile. The formation mean ChRM direction of the gabbro and the mean direction of the basic dyke is shown together with the 95 per cent confidence circle on top of the basic dyke. Negative inclinations are marked with open symbols.

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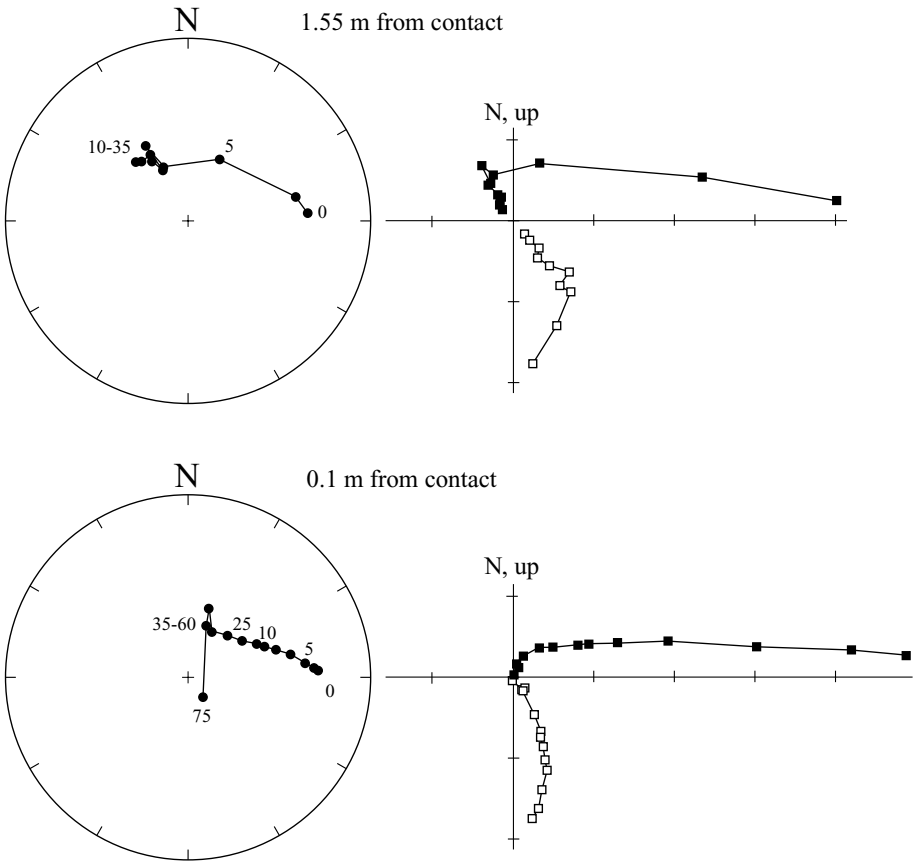


Figure 7. Examples of AF demagnetizations of gabbro samples collected in the hybrid zone and close to the contact of the dyke at site 3 (see Fig. 6). Note the pure gabbro direction isolated in alternating fields >10 mT in the hybrid zone, while the direction at fields >35 mT in the sample from the contact is a mixed dyke and gabbro direction.

ca. 5 min, whereas the time τ that the hybrid rock spent within 10 °C of its maximum reheating temperature is determined from eq. (2) as *ca*. 65–150 d. This is then used for the correction of the maximum temperature ($T_{max} \rightarrow T^*_{max}$) in the hybrid samples (Table 4).

CALCULATION OF THE AMBIENT TEMPERATURE AND UPLIFT

From eqs (1) and (2), ΔT_{max}, the contribution of the cooling dyke, and T_{amb}, the temperature of the host rock just prior to the dyke intrusion, have been calculated for each sample in the hybrid zone. The corrected values of T^*_{max}, ΔT^*_{max} and T^*_{amb} are listed in Table 4. Note that the estimated maximum temperature decreases with increasing distance from the dyke (Fig. 11). This, together with the facts that the magnetic properties in the hybrid zone are not different from those of the host rock and that the dyke component occupies a T_{ub} range below that of the host rock, supports a partial thermoremanent origin of the overprinting in the hybrid zone. The corrected ambient temperature (T^*_{amb}) ranges from 364 to 400 °C, with a mean value of 375 ± 10.7 °C (Table 4).

The present thermal gradient in the Hoting region is approximately 18 °C km^{-1} (Parasnis 1975). To estimate the thermal gradient at the time of the dyke intrusion, this gradient must be corrected for the decrease in heat production in the crust since that time. The heat is produced by radioactive decay of U, Th and K in various rocks in the crust. If the composition of the crust is simplified to be composed of an equal amount of granite, gabbro and sedimentary rocks, the heat production can be calculated from an average concentration of U, Th and K and the heat production of each element. The calculation is based on the main radioactive decay relation [$N = N_0 \exp^{(-\lambda t)}$] and yields a heat production at *ca*. 1.6 Ga, 1.9 times higher than at present, which corresponds to a palaeothermal gradient of 34 °C km^{-1}. Assuming a palaeosurface temperature of *ca*. 20 °C, which is reasonable since the estimated palaeolatitude for Fennoscandia at the time of intrusion is very close to the equator (*ca*. 4°N), the depth of burial of the present surface in the Hoting area at *ca*. 1.6 Ga is 10.4 km. If we assume an uncertainty of ±30 per cent in the estimated gradient (Parasnis 1975; Schwarz *et al.* 1985), the error in the depth of burial would be 3.1 km. This depth of burial is not very different from that estimated on the basis of geological data at *ca*. 1.6 Ga for southwestern Sweden (Tullborg *et al.* 1996). The uplift rate of the Hoting region since *ca*. 1.6 Ga is on the basis of the calculated depth of burial 0.65 km (100 Ma)$^{-1}$.

DISCUSSION

The rock magnetic properties, the decrease in T_{max} with increasing distance from the dyke contact and the reliable results obtained from the palaeointensity study argue for the magnetization in the hybrid zone at site 3 to be pure TRM.

The two basic dykes of this study form part of a group of dykes identified in a palaeomagnetic study of the regional dyke swarm (Elming *et al.* 2004). This group of dykes has a chemical

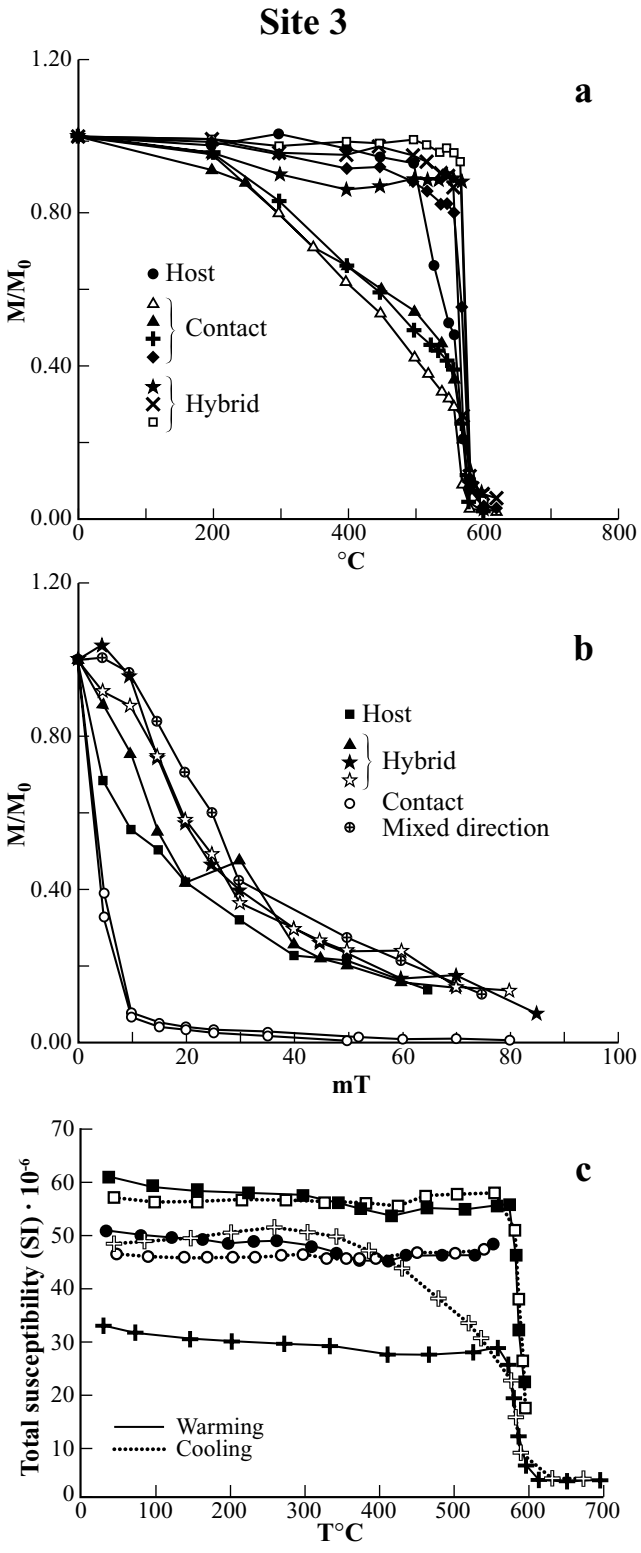


Figure 8. Thermal (a) and AF demagnetization (b) graphs of samples from the gabbro host rock, from the baked host rock and from the hybrid zone in the host rock. Note the similarity in unblocking temperature and coercivity spectra for the samples of the host rock and those from the hybrid zone, while the spectra for the baked host rock is different. Magnetic susceptibility variations (c) with increasing temperature of an unbaked gabbro sample from site 3. The three graphs represent heating to different maximum temperatures at 500, 600 and 700 °C, respectively.

composition that is similar to rapakivi related dykes in the Fennoscandian Shield. Rapakivi magmatism from the northwestern part of the shield is reported at ages of 1.51–1.58 Ga, which is similar to the calculated 1.6 Ga $^{40}Ar/^{39}Ar$ age of the basic dyke at site 3. The mean palaeomagnetic pole calculated for the group of dykes (Plat. = 21.9°N, Plon. = 146.7°E; Fig. 12 and Table 2) includes the site means 20a and 21 in Elming *et al.* (2004, in their table 1). This pole falls close to 1.578 Ga poles of the Nordingrå rapakivi granite and gabbro in central Sweden (Piper 1980). However, these poles are due to the slow cooling of rocks probably somewhat younger. The pole position calculated from the dyke suggests that Fennoscandia was located close to the equator at the time of the dyke intrusions.

Zircons were dated from three different samples of the gabbro (Hellström & Larson 2003), resulting in an upper and lower intercept U-Pb age of 1.786 ± 0.010 and 0.453 ± 0.019 Ga, respectively, where the lower intercept age might be related to Caledonian orogenic activity. With reference to the U-Pb age, the $^{40}Ar/^{39}$ biotite age (1.648 Ga) probably reflects the uplift and cooling of the gabbro. The palaeomagnetic pole calculated from this mean (Plat. = 43.0°N, Plon. = 233.3°E) falls very close to the 1.88 Ga reference pole for Fennoscandia (Plat. = 41.0°N, Plon. = 233°E; Fig. 12; Pesonen *et al.* 2003); however, the closing time of the palaeomagnetic system of the gabbro is different.

The closing temperature of the $^{40}Ar/^{39}Ar$ system for biotite is in the range 300–370 °C; (Berger & York 1981; Harrison *et al.* 1985), which means the host rock temperature at 1.648 ± 0.012 Ga, the biotite age of the pure gabbro, was not more than *ca*. 370 °C. This shall be compared with the host rock temperature 375 ± 10.7 °C estimated from the palaeomagnetic data for the time of the dyke intrusion at 1.614 ± 0.024 Ga. Considering the error bar of the palaeomagnetically determined host rock temperature and the overlapping error bars of the biotite ages, the calculated ambient temperature of 375 °C is not contradicted by the $^{40}Ar/^{39}Ar$ plateau ages of biotite. These ages probably relate to the uplift, with the biotite $^{40}Ar/^{39}Ar$ system closing shortly after the time at which the dyke intruded, probably slightly before 1.614 Ga ago.

Under the above condition, only one spot reading of the depth of burial of the area has been obtained. However, the contour map of the amount of uplift of the Fennoscandian Shield supports our results (Fjeldskaar 1994). The uplift rate of 0.65 km (100 Ma)$^{-1}$ can be compared with the rate of uplift of the Canadian Shield: 0.76 km (100 Ma)$^{-1}$ for the Sudbury dyke (1.25 Ga; Schwarz & Buchan 1982); 1.14 km/100 Ma for a Grenvillian diabase dyke (0.57 Ga; Hyodo *et al.* 1993) and 0.30 km (100 Ma)$^{-1}$ for the diabase dyke in the Munro Township of the Superior Province (2.15 Ga; Buchan & Schwarz 1981). The calculated uplift rate of the Hoting region thus falls well into the range of the uplift rate for the Canadian Shield and is close to that of the Sudbury dyke, which is of a similar age.

If temperatures of *ca*. 850–900 °C are estimated for the closing of the U-Pb in zircon (>900 °C for pristine zircon; Lee *et al.* 1997) and 300–370 °C for the $^{40}Ar/^{39}Ar$ of biotite systems (Berger & York 1981; Harrison *et al.* 1985), this would suggest a cooling rate of *ca*. 3.5 °C Ma^{-1}. A cooling rate of *ca*. 4.6 °C Ma^{-1} was suggested for a quartz syenite and granite of the Ntonya complex (Briden *et al.* 1993) and a rate of 1 °C Ma^{-1} was estimated for the cooling of a gabbro in the Grenvillian Province (Baksi 1982). This means that the assumption of slow cooling of the Hoting gabbro to explain the age difference between the U-Pb and Ar/Ar ages is reasonable.

From the estimated cooling rate of the gabbro (3.5 °C Ma^{-1}), the time for the closure of the palaeomagnetic system can be

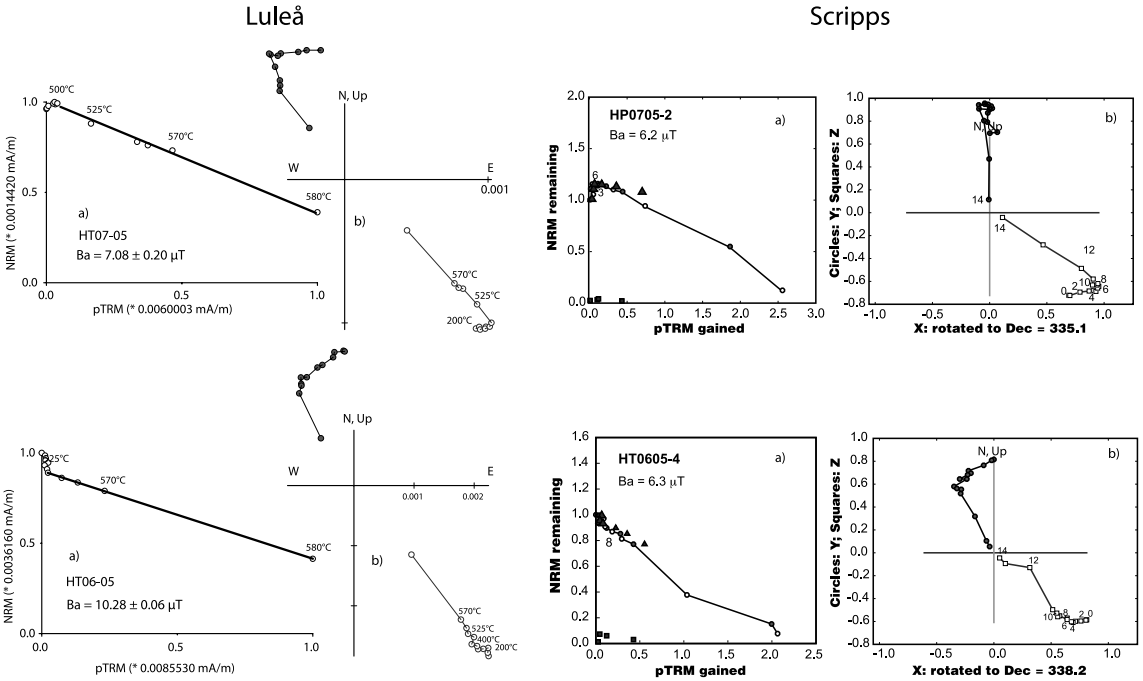


Figure 9. Examples of palaeointensity experiments with vector plots from two samples collected in the hybrid zone at site 3. The Arai plots from the Luleå laboratory are based on the Coe method (Coe 1967), while the results from Scripps laboratory are based on the IZZI method (Tauxe & Staudigel 2004). The determined ancient field Ba is also given. The open (closed) symbols in the vector plots denote inclinations (declination). Note that the palaeointensity results from the two methods are similar.

Table 3. Intensity results (B_a) from Scripps and Luleå.

Specimen	T_{min} (°C)	T_{max} (°C)	N	B_l (mT)	B_a (mT)	σB_a	q	f (coe)	DRATS (per cent)	MD (per cent)	D (°)	I (°)	MAD/α_{95} (°)	VDM (10^{22} Am^2)	σVDM
Scripps															
HP0605-1	560	581	6	15	4.3		8.2	0.83	12.0	2	328.6	36.1	6.0		
HP0605-2	560	581	6	15	4.4		9.9	0.79	13.5	5	308.2	31.3	8.0		
HP0605-3	560	581	6	15	3.2		7.3	0.81	15.6	5	306.9	34.8	11.9		
HP0605-4	550	581	7	15	6.3		7.5	0.85	18.9	4	308.9	42.6	7.9		
HP0605-5	550	581	7	15	6.4		10.0	0.84	22.6	5	315.4	38.2	11.1		
HP0705-1	550	581	7	15	5.0		10.4	0.84	21.9	3	330.3	44.8	3.4		
HP0705-2	550	581	7	15	6.2		10.3	0.82	17.2	2	331.1	35.3	5.7		
HP0705-3	550	581	7	15	5.6		9.2	0.81	25.3	6	336.0	34.4	5.1		
HP0705-4	550	581	7	15	5.0		8.1	0.81	32.0	2	335.1	39.0	8.0		
Luleå															
HT07-05	525	580	5	50	7.1		6.6	0.49			326.1	44.6	5.9		
HT06-05	525	580	5	50	10.3		35.9	0.53			331.3	51.7	2.4		
Mean Scripps			9		5.2	1.1								0.90	0.19
Mean DRATS <20 per cent			5		4.9	1.3								0.88	0.23
Mean Luleå			2		8.7	–								1.50	–
Mean all			11		5.8	1.9					323.2	39.9	6.0	1.00	0.32

Notes: The table shows the temperature interval (T_{min} to T_{max}) used for calculation of the palaeointensity, which is based on the stability of the direction (expressed as maximum angular deviation, MAD). The number of points used for the calculation of the slope (N), the laboratory field (B_l), the determined palaeofield field (B_a), the quality factor (q), fraction of NRM (f), DRATS, and tail checks (MD per cent) are also indicated. D and I stands for declination and inclination, respectively, of the erased remanence vector. The virtual dipole moment (VDM) is calculated for the mean intensities. σB_a and σVDM denote the standard deviation of the calculated mean palaeofield and the VDM, respectively.

calculated. The ChRM in the gabbro is defined in a T_{ub} range of *ca*. 550–570 °C, which, as this is close to the Curie temperature of magnetite, is similar to the natural T_b for such a cooling rate (Dodson & McClelland-Brown 1980). This means that the closing time for the palaeomagnetic system can be estimated at *ca*. 1.7 Ga, a time at which the intensity of the field and the VDM is here estimated at 5.8 ± 1.9 μT and 1.0 × 10^{22} Am^2, respectively. The slow cooling of the gabbro suggests that periodical changes in the

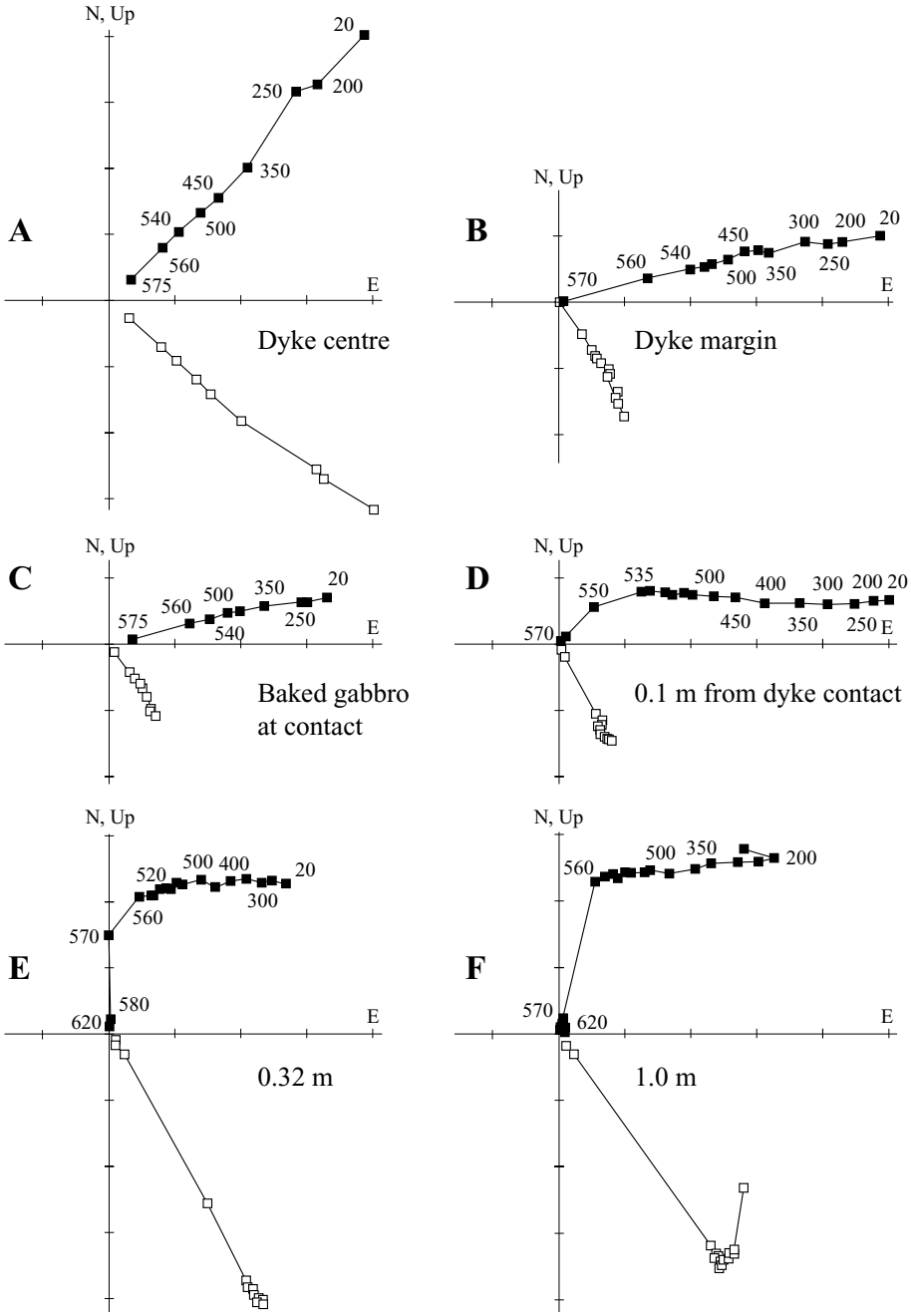


Figure 10. Vector plots of samples collected from the basic dyke and from the host rock at increasing distance from the dyke at site 3. Peak temperatures are given in degrees Celsius. Open (closed) symbols in the vector plot indicate vertical (horizontal) projections. Note the gradually changing directions when moving away from the dyke contact and that the host rock direction (L) cannot be isolated in samples closer than 1.3 m. In the hybrid zone (1.3–2.28 m), both the dyke and the host rock directions can be defined and T_{max} is defined in samples from this zone. At for example, 1.46 m (H), T_{max} is defined from the average of 540 and 570 °C, that is, 555 °C.

intensity of the field in the timescale of *ca*. 100 000 kyr, as observed during the last *ca*. 800 000 kyr (e.g. Yamazaki & Oda 2002), may be averaged out and the estimated palaeointensity thus represents an average amplitude of the field. Palaeointensity data from the Precambrian are very sparse and scattered. Different models have been presented, for example, Merrill *et al.* (1996), in which the intensity is either higher or very much lower than the PEF. The low amplitude of the Earth's magnetic field during the Precambrian obtained here (Fig. 13) agrees with the results from, for example, *ca*. 2.2 Ga Proterozoic dykes from the Superior province (Macouin *et al.* 2003).

Even if the reliable palaeointensities further suggest that the TRM is of primary origin, one should keep in mind that the possibility of the remanence to be of thermochemical origin, which biases the

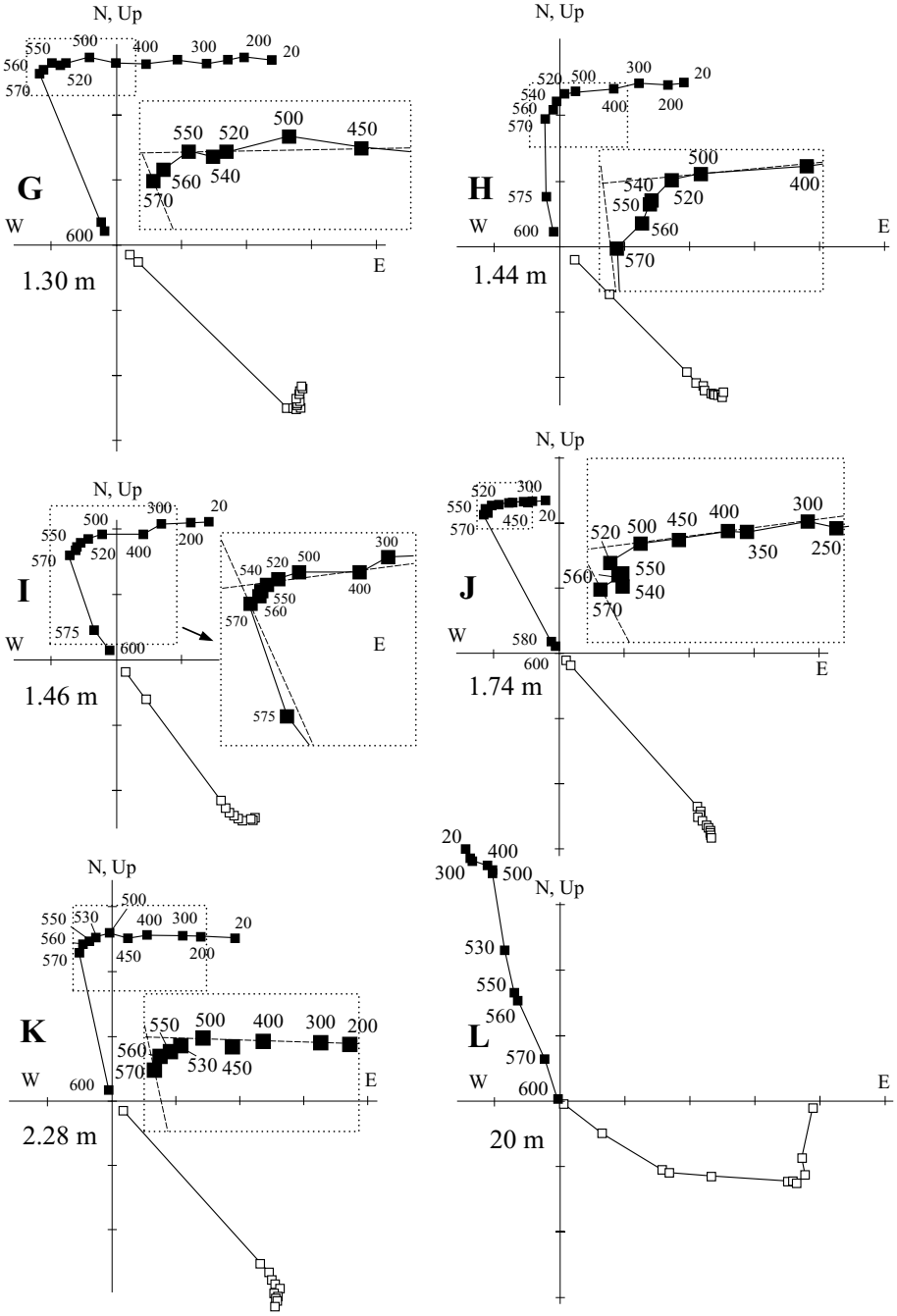


Figure 10. (*Continued*.)

results towards lower values, cannot be ruled out (e.g. Smirnov & Tarduno 2005). To answer this open question, petrographic analyses will be performed.

The calculated palaeomagnetic pole from the basic dykes fulfils most of the criteria for a reliable pole (Buchan *et al.* 2001) and may be regarded as a new key-pole for Fennoscandia. Also, the pole determined from the gabbro is reliable; however, there is no field test to demonstrate a primary origin of magnetization.

CONCLUSION

From the magnetic properties, the palaeomagnetic data and palaeointensity studies it can be concluded that the overprinting in the hybrid zone of the basic dyke at site 3 is a partial thermomagnetic magnetization. On basis of the maximum temperature determined in the hybrid zone the ambient temperature in the host rock at the time of the dyke intrusion, slightly before 1.614 Ga, is calculated at 375 °C. A palaeothermal gradient in the crust at *ca*. 1.6 Ga is estimated at *ca*. 34 °C km^{-1}, which yields a depth of burial of the present erosion surface of 10.4 km and an uplift rate of 0.65 km 100 Ma^{-1}. The cooling rate of the gabbro is calculated at *ca*. 3.5 °C Ma^{-1}, which suggests a palaeomagnetic age of 1.7 Ga for the pole of the gabbro. The intensity of the Earth's magnetic field at that time is determined at 5.8 $\pm$ 1.9 μT, which is very much lower than that of the present Earth field.

Table 4. T_{max} as estimated from the vector plots, that is, the average of the maximum temperature at which the dyke direction is identified and the minimum temperature at which the host rock gabbro direction is identified.

Distance from the contact (m)	T_{max}	ΔT_{max}	T_{amb}	T^*_{max}	ΔT^*_{max}	T^*_{amb}
1.3	550–570	180	380	555	177	378
1.3	560	180	380	555	177	378
1.3	540–570	181	374	550	183	367
1.38	550–560	173	382	545	175	370
1.44	520–570	169	376	535	171	364
1.45	530–560	168	377	535	170	365
1.46	540–570	165	390	545	167	378
1.65	520–570	151	394	535	153	382
1.74	500–570	146	389	520	149	371
2.28	500–570	116	419	518	118	400
Mean						375 ± 10.7

Notes: ΔT_{max} is the temperature increase due to the cooling dyke and T_{amb} is the calculated ambient temperature just prior to the dyke intrusion. T^*_{max}, ΔT^*_{max} and T^*_{amb} are the values corrected for viscous effects.

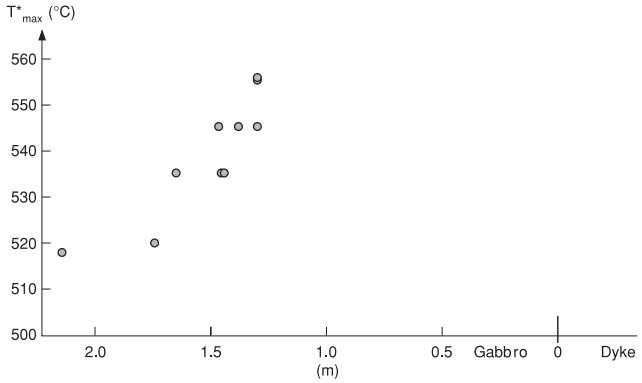


Figure 11. The maximum temperature T^*_{max} determined at increasing distance from the contact to the basic dyke. Note the gradual decrease in T^*_{max} with increasing distance.

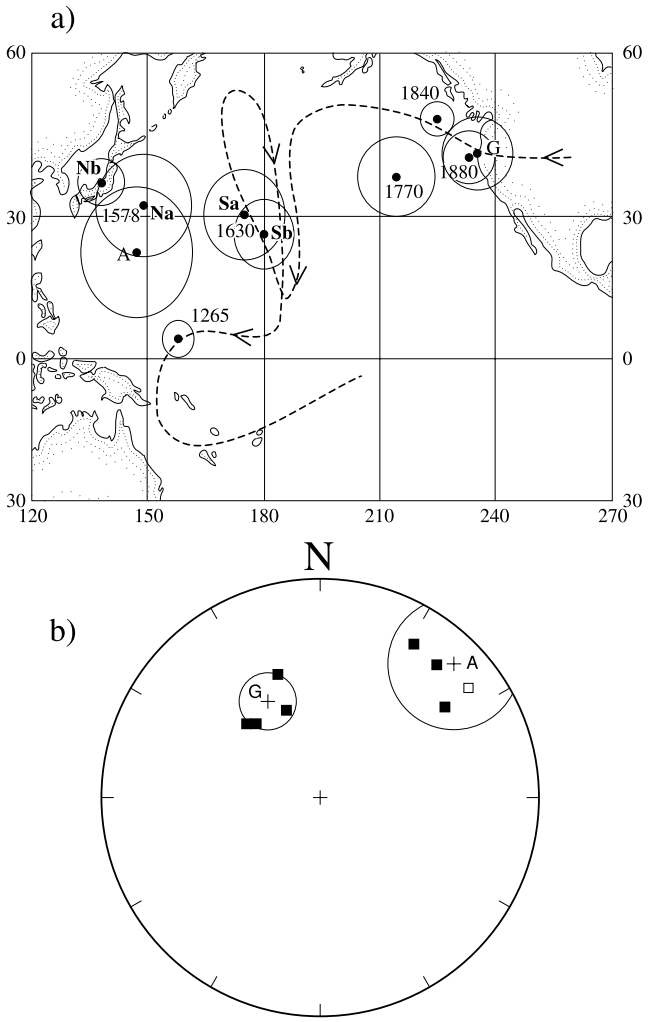


Figure 12. (a) Palaeomagnetic reference poles at 1.88, 1.84, 1.77, 1.63 and 1.265 Ga and the Apparent Polar Wander Path for Fennoscandia (Elming *et al.* 1993; Pesonen *et al.* 2003). The poles A and G are calculated from the formation mean directions (b) of the basic dykes and the gabbro, respectively, of this study. The 1.578 Ga poles are from the Nordingrå rapakivi granite (Na) and gabbro-anorthosite (Nb) (Piper 1980) and the 1.630 Ga poles from the SE-quartz porphyry dykes (Sa; Neuvonen 1986) and the Sipoo quartz porphyry dykes (Sb; Mertanen & Pesonen 1995).

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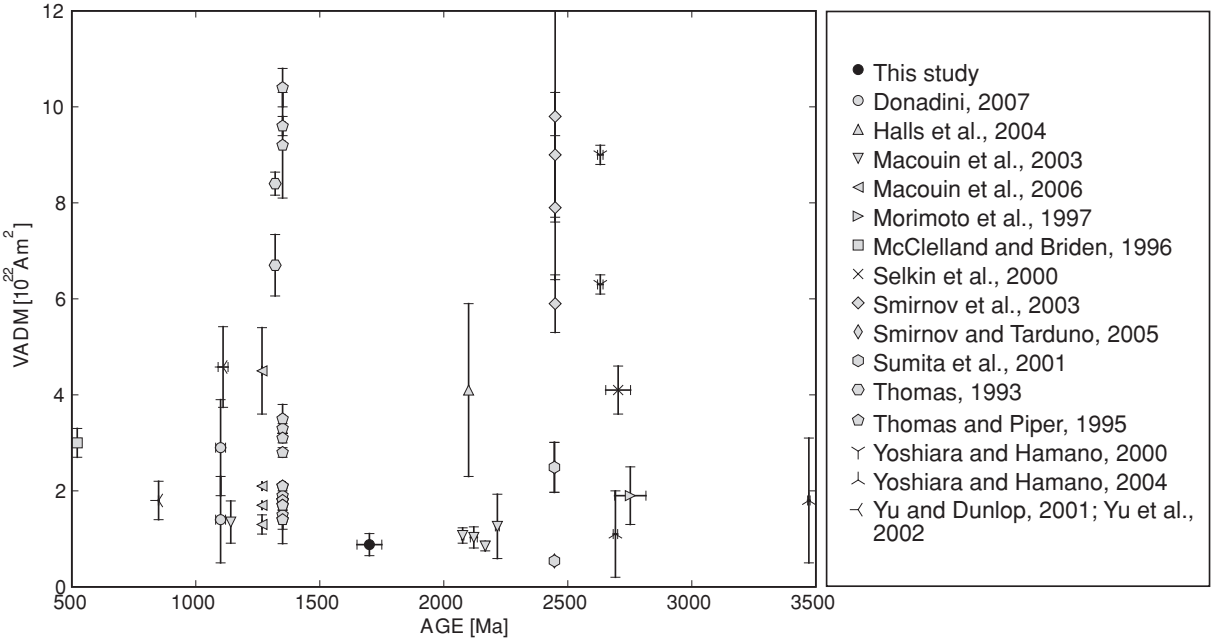


Figure 13. Precambrian palaeointensity data with error bars presented as VDM values. Note that our new data (filled star) is plotted in a big gap of data and indicates a very low intensity of the palaeofield.

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APPENDIX

A.1 Analytical data

Weighted average J (irradiation parameter) calculated from standard MMhb-1 (513.9 Ma) Runs are step heat analyses of single crystals of a mineral phase or ∼1 mm whole-rock chips. Laser power (in milliwatts) is the heating step from a defocused argon-ion laser: 8700 mW represents the fusion step in most cases per cent ^{39}Ar: the proportion of ^{39}Ar released in each step of a run Measured isotopic ratios (and 1σ error) are corrected for reactor induced interferences and decay of ^{37}Ar and ^{39}Ar per cent Atm. ^{40}Ar: per cent of non-radiogenic ^{40}Ar in the sample assuming an initial ^{40}Ar/^{36}Ar ratio of 295.5 $^{40}Ar^{*}/^{39}$ArK and ages (and 1σ errors) calculated using the equations and constants quoted in McDougall Harrison (1999) Bold steps represent those used to calculate plateaus (see text and Fig. 2).

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H5.94 Site 1 Biotite Weighted average of J from standards = 0.002402 +/− 0.000012

Laser (mW)	Cum. ^{39}Ar	^{40}Ar/^{39}Ar measured	+/−	^{37}Ar/^{39}Ar measured	+/−	^{36}Ar/^{39}Ar measured	+/−	per cent Atm. ^{40}Ar	Ca/K	Cl/K	^{40}Ar*/^{39}ArK	+/−	Age (Ma)	+/− (Ma)
200	0.011	298.6	36.8	−0.101	0.262	0.012	0.133	1.2	−0.18	0.019	294.95	53.53	966.3	135.8
400	0.098	583.9	12.2	−0.026	0.028	−0.002	0.017	−0.1	−0.05	0.006	584.35	13.14	1582.3	23.7
600	**0.375**	**624.4**	**6.1**	**0.006**	**0.015**	**−0.006**	**0.006**	**−0.3**	**0.01**	**−0.001**	**626.07**	**6.37**	**1655.9**	**11.0**
800	**0.726**	**612.1**	**13.2**	**0.025**	**0.008**	**0.006**	**0.005**	**0.3**	**0.05**	**0.002**	**610.39**	**13.29**	**1628.6**	**23.4**
1000	**0.813**	**600.7**	**12.3**	**−0.032**	**0.035**	**−0.013**	**0.017**	**−0.7**	**−0.06**	**0.003**	**604.55**	**13.32**	**1618.3**	**23.6**
1200	**0.871**	**664.2**	**26.5**	**0.055**	**0.060**	**−0.008**	**0.029**	**−0.3**	**0.10**	**−0.001**	**666.51**	**27.91**	**1724.6**	**46.5**
1500	**0.942**	**637.1**	**14.0**	**0.007**	**0.061**	**−0.016**	**0.025**	**−0.7**	**0.01**	**0.002**	**641.74**	**15.90**	**1682.9**	**27.1**
2000	0.963	695.4	41.6	−0.137	0.142	0.044	0.084	1.9	−0.25	0.019	682.32	47.70	1750.8	78.3
2500	0.966	811.7	355.4	−0.468	1.122	0.105	0.628	3.8	−0.86	0.010	780.44	388.48	1905.1	585.7
3000	0.967	3546.0	10514.2	1.585	7.578	7.798	23.354	65.0	2.91	0.625	1243.17	3816.45	2494.9	4149.1
4000	0.992	407.0	29.7	2.443	0.225	1.790	0.147	129.9	4.49	0.164	−121.91	22.19	−625.1	136.0
8700	1.000	2828.0	470.6	2.216	0.620	8.786	1.506	91.8	4.07	0.138	232.17	121.56	799.6	338.2
Integrated		629.4	5.9	0.084	0.012	0.118	0.006	5.5	0.15	0.008	594.54	6.10	1600.5	12.1

H30.95 Site 3 Biotite Weighted average of J from standards = 0.002402 +/− 0.000012

Laser (mW)	Cum. ^{39}Ar	^{40}Ar/^{39}Ar measured	+/−	^{37}Ar/^{39}Ar measured	+/−	^{36}Ar/^{39}Ar measured	+/−	per cent Atm. ^{40}Ar	Ca/K	Cl/K	^{40}Ar*/^{39}ArK	+/−	Age (Ma)	+/− (Ma)
200	**0.254**	**588.5**	**17.5**	**0.009**	**0.067**	**−0.003**	**0.033**	**−0.2**	**0.02**	**0.001**	**589.39**	**20.10**	**1591.3**	**36.1**
400	**0.406**	**608.8**	**35.0**	**0.123**	**0.109**	**−0.048**	**0.052**	**−2.3**	**0.22**	**0.004**	**623.05**	**38.97**	**1650.7**	**67.7**
600	**0.701**	**602.8**	**15.2**	**0.032**	**0.063**	**−0.008**	**0.021**	**−0.4**	**0.06**	**0.003**	**605.24**	**16.47**	**1619.5**	**29.1**
800	**0.846**	**605.6**	**36.7**	**−0.105**	**0.094**	**−0.052**	**0.036**	**−2.6**	**−0.19**	**0.009**	**621.05**	**39.17**	**1647.2**	**68.1**
1000	**0.891**	**509.3**	**88.4**	**0.335**	**0.324**	**−0.037**	**0.205**	**−2.2**	**0.61**	**0.022**	**520.36**	**108.74**	**1463.0**	**209.4**
1200	0.909	294.8	150.0	−0.334	0.934	−0.135	0.504	−13.5	−0.61	−0.101	334.40	225.21	1063.7	541.3
1500	0.934	140.9	48.7	0.461	0.626	−0.298	0.322	−62.5	0.85	0.032	229.07	119.93	791.0	335.3
2500	0.942	171.3	174.0	1.371	2.498	−0.223	0.835	−38.5	2.52	−0.107	237.35	338.66	814.0	934.7
3000	0.955	75.7	58.6	0.999	1.429	−0.467	0.825	−182.5	1.83	0.199	213.89	274.76	748.0	786.6
4000	0.966	107.2	97.8	−1.714	2.358	−1.422	1.534	−392.1	−3.14	0.038	526.63	537.26	1475.1	1028.0
8700	1.000	430.1	110.1	−0.871	0.582	−0.096	0.210	−6.6	−1.60	−0.071	458.23	132.52	1339.2	273.4
Integrated		557.8	16.1	0.012	0.055	−0.056	0.026	−3.0	0.02	0.003	574.33	18.26	1564.1	33.7

H30.95 Site 3 plagioclase Weighted average of J from standards = 0.002402 +/− 0.000012

Laser (mW)	Cum. ^{39}Ar	^{40}Ar/^{39}Ar measured	+/−	^{37}Ar/^{39}Ar measured	+/−	^{36}Ar/^{39}Ar measured	+/−	per cent Atm. ^{40}Ar	Ca/K	Cl/K	^{40}Ar*/^{39}ArK	+/−	Age (Ma)	+/− (Ma)
1000	0.188	21.2	9.6	12.863	3.756	−0.063	0.189	−91.9	23.80	0.037	40.95	57.88	169.3	228.4
1200	0.313	238.6	84.5	19.687	7.030	−0.020	0.506	−3.1	36.59	−0.016	249.32	175.88	846.7	476.7
1500	0.478	159.2	53.8	5.422	1.964	0.093	0.392	17.0	9.98	0.001	132.63	124.23	499.0	408.3
2000	**0.595**	**148.0**	**50.6**	**5.676**	**2.264**	**−0.396**	**0.637**	**−79.4**	**10.45**	**0.100**	**266.57**	**205.97**	**892.9**	**544.2**
2500	**0.683**	**30.0**	**27.4**	**2.532**	**2.784**	**−1.616**	**1.617**	**−1594.0**	**4.65**	**0.122**	**508.67**	**504.02**	**1440.4**	**983.1**
3000	**0.791**	**−10.0**	**6.5**	**3.653**	**2.407**	**−0.432**	**0.766**	**1275.3**	**6.72**	**0.155**	**118.11**	**225.51**	**450.6**	**761.3**
4000	**0.860**	**3.0**	**6.0**	**4.960**	**4.500**	**−0.144**	**1.059**	**−1443.4**	**9.13**	**0.003**	**45.98**	**314.26**	**189.0**	**1226.5**
8700	**1.000**	**342.1**	**121.9**	**18.034**	**6.474**	**0.040**	**0.545**	**3.0**	**33.48**	**−0.030**	**335.68**	**202.89**	**1066.8**	**486.8**
Integrated		2.6	4.3	11.528	2.506	−0.591	0.256	−6918.5	21.31	−0.004	179.32	76.81	646.3	232.7

H30.95 Site 3 whole rock Weighted average of J from standards = 0.002402 +/− 0.000012

Laser (mW)	Cum. ^{39}Ar	^{40}Ar/^{39}Ar measured	+/−	^{37}Ar/^{39}Ar measured	+/−	^{36}Ar/^{39}Ar measured	+/−	per cent Atm. ^{40}Ar	Ca/K	Cl/K	^{40}Ar*/^{39}ArK	+/−	Age (Ma)	+/− (Ma)
200	**0.056**	**137.3**	**91.1**	**−0.974**	**1.869**	**−0.376**	**0.852**	**−80.8**	**−1.79**	**0.032**	**248.06**	**291.42**	**843.3**	**791.4**
400	**0.235**	**208.5**	**36.6**	**3.165**	**0.839**	**−0.071**	**0.264**	**−10.1**	**5.82**	**0.014**	**230.04**	**87.83**	**793.7**	**245.2**
600	**0.439**	**211.3**	**34.0**	**1.514**	**0.635**	**−0.011**	**0.225**	**−1.6**	**2.78**	**−0.035**	**214.90**	**74.98**	**750.9**	**214.3**
800	**0.505**	**559.9**	**332.7**	**6.778**	**4.303**	**−0.534**	**0.713**	**−28.3**	**12.49**	**−0.100**	**721.30**	**470.45**	**1813.7**	**746.1**
1000	**0.587**	**218.4**	**95.6**	**4.437**	**2.368**	**0.356**	**0.568**	**48.1**	**8.16**	**−0.096**	**113.66**	**169.46**	**435.5**	**576.9**
1200	**0.640**	**237.0**	**170.5**	**8.575**	**6.379**	**0.858**	**0.877**	**106.7**	**15.82**	**0.093**	**−15.99**	**185.60**	**−70.7**	**836.5**
1500	**0.686**	**591.0**	**478.0**	**28.412**	**23.061**	**0.648**	**0.918**	**32.0**	**53.11**	**0.779**	**409.37**	**406.63**	**1235.5**	**888.5**
2000	**0.774**	**187.0**	**78.7**	**7.992**	**3.489**	**−0.025**	**0.339**	**−4.2**	**14.74**	**0.209**	**195.85**	**130.50**	**695.7**	**384.6**
2500	0.836	112.6	76.7	4.842	3.557	0.346	0.630	90.6	8.91	0.066	10.67	173.45	45.7	732.9
3000	0.928	84.5	32.2	5.213	2.141	0.592	0.455	206.7	9.60	0.080	−90.43	122.21	−441.8	676.6
4000	0.987	23.3	15.1	1.462	1.791	0.515	0.706	654.4	2.69	0.060	−129.00	202.32	−669.0	1270.6
8700	1.000	922.1	2738.6	49.495	147.131	2.251	7.150	71.7	93.84	0.863	269.27	1132.11	900.0	2979.3
Integrated		227.9	29.3	5.641	0.796	0.168	0.138	21.6	10.39	0.068	179.30	46.51	646.2	140.9

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H44.94 Basic dyke whole rock Weighted average of J from standards = 0.002402 +/− 0.000012

Laser (mW)	Cum. ^{39}Ar	^{40}Ar/^{39}Ar measured	+/−	^{37}Ar/^{39}Ar measured	+/−	^{36}Ar/^{39}Ar measured	+/−	per cent Atm. ^{40}Ar	Ca/K	Cl/K	^{40}Ar*/^{39}ArK	+/−	Age (Ma)	+/− (Ma)
200	0.011	281.1	239.5	1.618	2.246	−0.219	0.807	−23.1	2.97	0.355	346.40	375.72	1092.3	888.8
400	0.078	152.7	20.4	0.681	0.199	−0.306	0.173	−59.3	1.25	0.006	243.23	59.50	830.1	162.8
600	0.145	362.1	50.8	0.764	0.240	−0.136	0.141	−11.1	1.40	0.042	402.49	69.90	1220.4	154.0
800	**0.316**	**468.8**	**25.8**	**0.354**	**0.076**	**−0.006**	**0.061**	**−0.4**	**0.65**	**0.008**	**470.64**	**31.60**	**1364.6**	**64.3**
1000	**0.495**	**522.7**	**30.7**	**1.253**	**0.111**	**−0.071**	**0.050**	**−4.0**	**2.30**	**−0.003**	**544.14**	**35.19**	**1508.2**	**66.1**
1200	**0.695**	**457.8**	**31.1**	**3.459**	**0.263**	**0.040**	**0.046**	**2.5**	**6.36**	**0.004**	**447.33**	**33.38**	**1316.6**	**69.7**
1500	**0.883**	**454.2**	**32.9**	**5.172**	**0.399**	**0.009**	**0.044**	**0.5**	**9.52**	**0.003**	**453.46**	**35.45**	**1329.3**	**73.5**
2000	**0.940**	**372.9**	**76.8**	**4.174**	**1.029**	**0.115**	**0.159**	**9.1**	**7.68**	**−0.002**	**340.03**	**84.30**	**1077.2**	**201.1**
2500	0.960	513.1	338.6	15.576	10.316	−0.235	0.520	−13.8	28.87	−0.028	589.72	420.17	1591.9	753.5
3000	0.980	236.0	134.0	8.642	5.026	0.540	0.534	67.4	15.95	−0.044	77.37	137.19	307.5	501.4
4000	0.991	334.5	375.1	10.738	12.161	0.334	0.867	29.3	19.84	−0.160	238.24	355.68	816.5	980.4
8700	1.000	532.5	689.1	23.357	30.323	1.310	1.966	72.4	43.52	0.108	149.50	357.58	553.6	1140.2
Integrated		432.9	17.1	3.116	0.142	−0.008	0.032	−0.6	5.73	0.007	436.28	19.75	1293.3	42.0

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